

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2023 - 2024)

JEE(Advanced)**PRE-SCORE****03-12-2023****JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS, TAAS & TTAS)**

Time : 3 Hours

PAPER-2 (OPTIONAL)

Maximum Marks : 180

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their Form No. in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

READ THE INSTRUCTIONS CAREFULLY**GENERAL :**

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **17** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

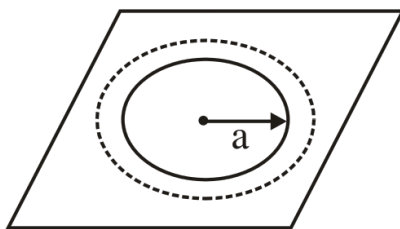
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. A uniform ring of mass m , radius a , resistance R , modulus of elasticity Y , area of cross section A is kept on a smooth surface. When it is somehow expanded & released, it undergoes radial oscillations with angular frequency ω_0 . If the same ring is made to oscillate radially in presence of vertical magnetic field B , it undergoes damped oscillation with angular frequency ω .



(A) $\omega_0 = \sqrt{\frac{4\pi Y A}{am}}$

(B) $\omega = \sqrt{\frac{4\pi Y A}{ma} - \frac{2\pi^4 a^4 B^4}{m^2 R^2}}$

(C) $\omega_0 = \sqrt{\frac{2\pi Y A}{am}}$

(D) $\omega = \sqrt{\frac{2\pi Y A}{ma} - \frac{2\pi^4 a^4 B^4}{m^2 R^2}}$

2. The pitch of screw gauge is 1 mm and its circular scale is divided into 100 divisions. When nothing is put between the studs the circular scale lies between 2nd and 3rd division of main scale, but when circular scale is rotated by 270° the 3rd division of main scale is seen and the 3rd division of main scale coincide with the 12th division of circular scale. When a glass plate is placed between the studs, the circular scale lies between 28th to 29th division of main scale and circular scale reads 20 divisions. Find the thickness of the glass plate in millimeter. (Assume value of one main scale division is same as pitch)

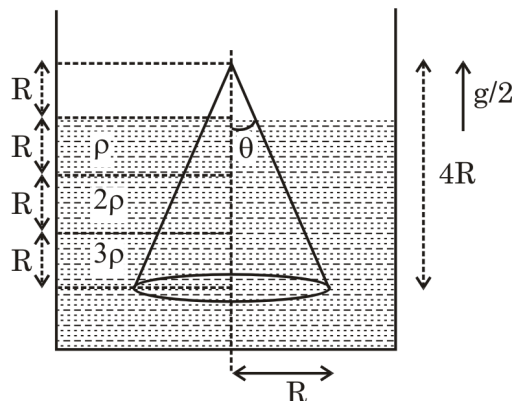
(A) 25.83 mm

(B) 26.83 mm

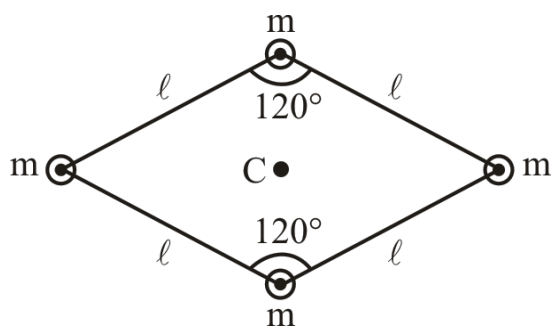
(C) 25.73 mm

(D) 25.80 mm

3. A solid cone is kept in three immiscible liquids of density ρ , 2ρ and 3ρ inside a container. The radius of cone is 'R' and it's height is $4R$. Container is accelerated with a constant acceleration $g/2$ in vertically upward direction as shown. If cone is in equilibrium relative to the frame of container then density of the cone is



- (A) $\frac{73}{24}\rho$ (B) $\frac{146}{25}\rho$ (C) $\frac{39}{16}\rho$ (D) $\frac{146}{37}\rho$
4. Consider four point masses each of mass m are fixed at the vertex of a Rhombus of side ℓ . The escape speed from the centre of the Rhombus is (G is universal gravitational constant)



- (A) $\sqrt{\frac{8Gm}{\ell}}$ (B) $\sqrt{4\sqrt{2}\frac{Gm}{\ell}}$
- (C) $\sqrt{\frac{8Gm}{\ell}\left(1 + \frac{1}{\sqrt{3}}\right)}$ (D) $\sqrt{\frac{8Gm}{\ell}\left(1 + \frac{1}{\sqrt{5}}\right)}$

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

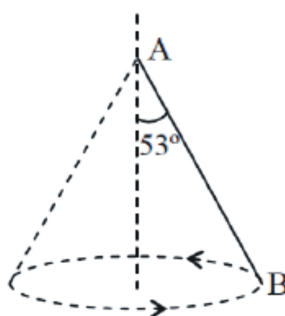
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. A uniform rod of length 4 m and mass $2\sqrt{2}$ kg revolves with constant angular velocity ω about a vertical axis through a smooth joint A at one extremity of the rod so that it describes a cone of semi vertical angle 53° as shown in the figure. Choose the correct statement(s). ($g = 10 \text{ m/s}^2$)

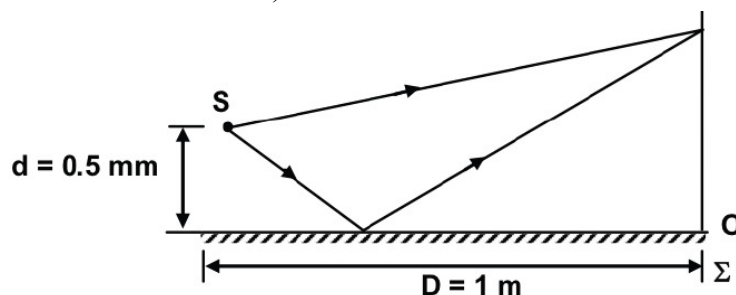


- (A) The value of angular velocity ω is 2.5 rad/s
- (B) The angle between hinge reaction at point A and vertical axis AC is 45°
- (C) The magnitude of hinge reaction at point A is 40 N
- (D) The angular momentum of the rod is constant

6. An imaginary particle has a charge equal to the charge of electron but mass of imaginary particle is 50 times the mass of electron. Now, consider that imaginary particle moves in circular orbit around a nucleus. The charge of nucleus is $+4e$, and nucleus is assumed to be at rest. Assume mass of nucleus is very large as compared to the mass of imaginary particle moving around it. Choose the correct option(s).

- (A) The radius of n^{th} orbit is $\frac{n^2 h^2 \epsilon_0}{200 \pi m e^2}$, where m is mass of electron, h is Planck constant, e is charge of electron.
- (B) If all atoms are at 16^{th} excited state then during de-excitation till 10^{th} excited state the total maximum number of possible wavelengths emitted is 21.
- (C) The wavelength of photon emitted when transition takes place from third excited state to ground state is $1.21 \text{ \AA}$.
- (D) Since mass of imaginary particle is different from the mass of electron so the velocity of imaginary particle and the electron in n^{th} orbit will be different.

7. A narrow slit S transmitting light of wavelength λ is placed a distance d above a large plane mirror as shown in the figure. The light coming directly from the slit and that coming after reflection from the plane mirror interfere on a screen Σ placed at a distance D from the slit. Assume that the mirror is 100% reflecting. (Given $D = 1 \text{ m}$, $d = 0.5 \text{ mm}$ and $\lambda = 500 \text{ nm}$)



- (A) Distance from O , where closest interference maximum is formed on the screen is $\frac{1}{4} \text{ mm}$.
- (B) Distance from O , where closest interference minimum is formed on the screen is $\frac{1}{4} \text{ mm}$.
- (C) Fringe width of the interference pattern formed on the screen is $\frac{1}{2} \text{ mm}$.
- (D) Closest point to O on the screen where waves interfere with a phase difference of π is $\frac{1}{8} \text{ mm}$.

SECTION-II (i) : (Maximum Marks: 12)

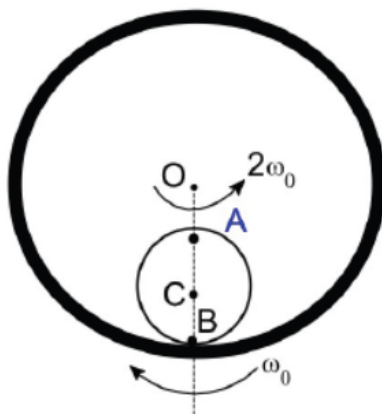
- This section contains **TWO (02)** paragraphs.
- Based on each paragraph, There are **TWO (02)** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

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Paragraph for Questions 1 and 2

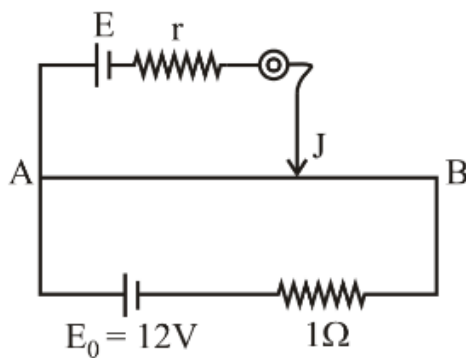
A hollow cylinder of radius $4R$ is rotating about fixed horizontal axis passing through point 'O' with angular velocity ω_0 . A solid cylinder of radius ' R ' is rolling without slipping with respect to inner surface of hollow cylinder. At the given instant, the line OC has angular velocity of $2\omega_0$. Point A and B are topmost and bottom most point of solid cylinder respectively and C is its centre. Then, at the given instant



1. Angular velocity of solid cylinder about its own axis is $\alpha \omega_0$, the value of α is :-
2. The magnitude of acceleration of point B of solid cylinder with respect to the contact surface of hollow cylinder is $\beta \omega_0^2 R$, the value of β is :-

Paragraph for Questions 3 and 4

In the set-up shown 'E' is unknown emf with internal resistance 'r'. AB is a uniform potentiometer wire, 11m long and resistance $1\Omega\text{m}^{-1}$. When $AJ = 1\text{m}$, current through galvanometer is 2.4 A when $AJ = 10\text{ m}$, current through galvanometer is $\frac{6}{11}\text{A}$, in the opposite direction as in previous case. Galvanometer has negligible resistance.



3. The value of E is (in V)
4. The value of r is (in Ω)

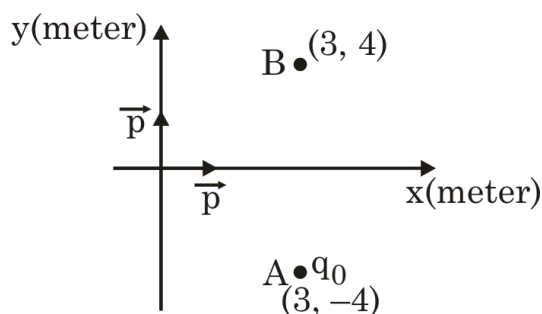
SECTION-II (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

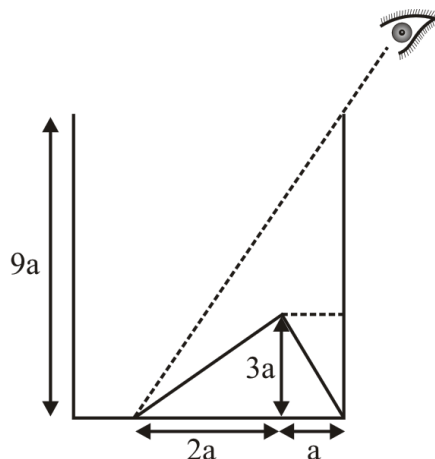
Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

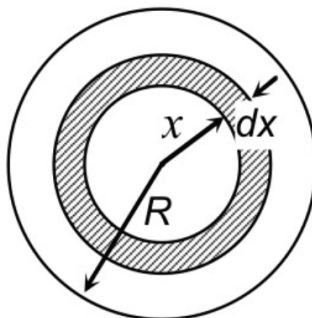
5. A particle of mass $m = 9 \text{ gm}$ is trapped between two perfectly rigid parallel walls. The particle bounces back and forth between the walls without losing any energy. From a wave point of view, the particle trapped between the walls is like a standing wave in stretched string between the walls. Distance between the two walls is $L = 1 \text{ m}$. Calculate the energy difference between third energy state and the ground state (lowest energy state) of the particle. Answer the energy difference in $C \times 10^{-65} \text{ Joule}$. (Take Planck's constant $h = 6 \times 10^{-34} \text{ J-sec}$)
6. Two short electric dipoles each of dipole moment P are placed at origin with direction along the coordinate axis as shown in the figure. Calculate the work done (in joule) in moving a point charge q_0 from coordinate $(3, -4)$ to $(3, 4)$? (Take : $\frac{Pq_0}{4\pi\epsilon_0} = 125 \text{ SI unit}$)



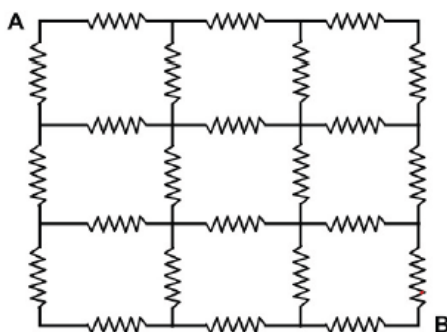
7. In a container of depth '9a' a triangular wedge is kept touching a wall as shown such that its one edge is just visible to an external observer. When a liquid of refractive index μ is completely filled in the container top of wedge also becomes visible. If the refractive index of liquid is $\sqrt{\frac{k}{10}}$, the value of k is :-



8. A string of 0.2 kg/m and length 1.2 m is fixed at both ends are stretched such that tension generated in the string is 320 N . The string vibrates in six segments with amplitude of 0.25 cm . Then maximum transverse velocity of a particle which is at a distance of $\frac{1}{30} \text{ m}$ from the fixed end is $k\pi \text{ cm/sec}$. Write value of k .
9. A uniform disc of radius R having charge Q distributed uniformly all over its surface is placed on a smooth horizontal surface. A magnetic field, $B = kxt^2$, where k is a constant, x is the distance (in metre) from the centre of the disc and t is the time (in second), is switched on perpendicular to the plane of the disc. Find the torque (in N-m) acting on the disc after 15 sec . (Take $4kQ = 5 \text{ S.I. unit}$ and $R = 1 \text{ m}$).



10. 24 resistors of 7Ω are arranged in a square as shown. Equivalent resistance of the network between points A and B is :-



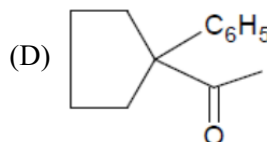
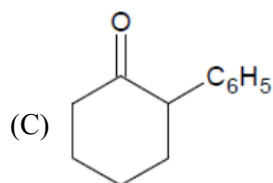
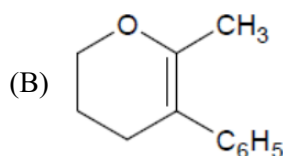
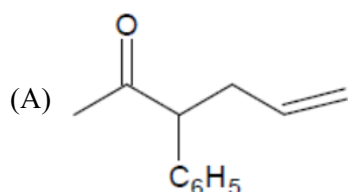
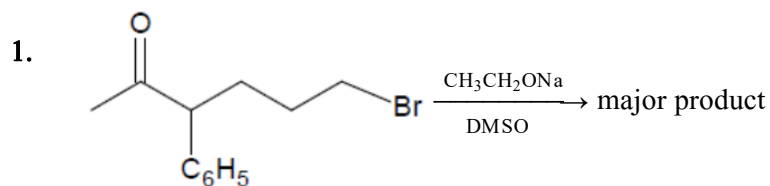
PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

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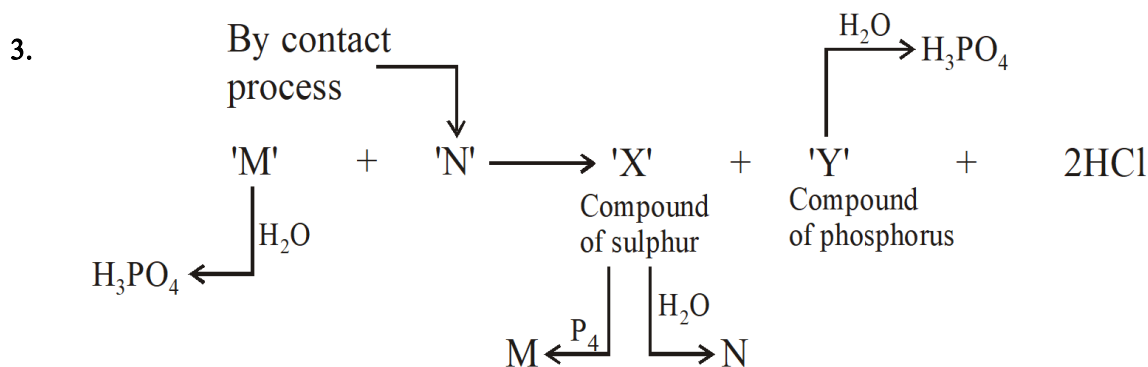
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases



2. A plot for $\log \frac{x}{m}$ vs $\log P$ shows a straight line with an angle 60° and intercept of 0.3010 on y-axis. The amount of gas adsorbed per gram of adsorbent if initial pressure maintained is 1 atm is:

- (A) 1 gm
(B) 2 gm
(C) 3 gm
(D) 4 gm



Select correct option regarding compound 'X' and 'Y' :

- (A) 'X' has sp^3d hybridisation but 'Y' has sp^3 .
- (B) Both 'X' and 'Y' have same total number of $p\pi - d\pi$ bonds.
- (C) Individual hydrolysed solution of 'X' and 'Y' converts blue litmus paper to red.
- (D) Both compound are planar.
4. Which of the following statement is **INCORRECT** ?
- (A) Cassiterite ore of tin contains the impurities of Woll-framite which are separated by magnetic separator.
- (B) Tin metal is obtained by the carbon reduction of concentrated ore.
- (C) In the extraction of lead from galena, the roasting and self-reduction are carried in the same furnace at different temperatures.
- (D) Primary reducing agent of haematite in blast-furnace is coke in upper part and CO in lower part of furnace.

SECTION-I (ii) : (Maximum Marks: 12)

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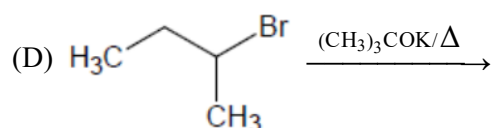
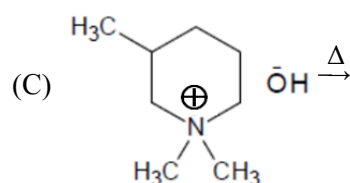
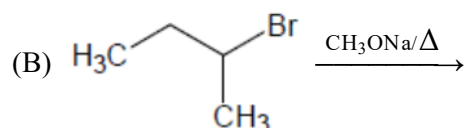
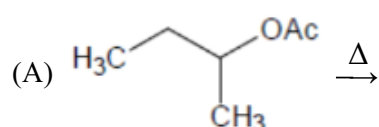
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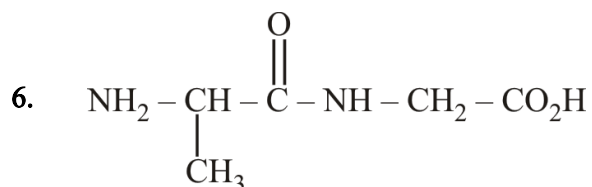
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5. In which of the following reaction product formation takes place by Hofmann rule?





Identify the amino acids obtained by hydrolysis of the above compound :

- (A) Glycine (B) Phenylalanine
(C) Alanine (D) Glutamic acid
7. A d-block metal ion with d^n configuration, when it forms tetrahedral complex but its spin magnetic moment remains same either with $4\text{H}_2\text{O}$ or with 4Cl^- ligand, which of the following statements are INCORRECT ?
- (A) Element always form a colourless compound
(B) Number of electrons in e-set of d-orbitals are always higher than in t₂-set of d-orbitals
(C) It can have d^5 or d^8 configuration
(D) Colour of the complexes must be same

SECTION-II (i) : (Maximum Marks: 12)

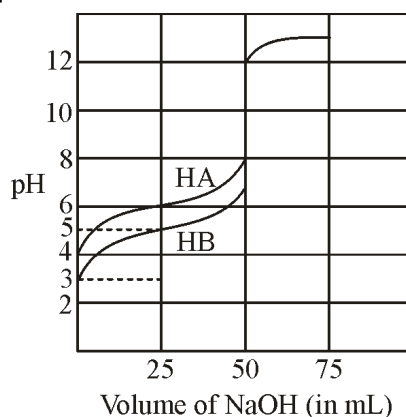
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Paragraph for Questions 1 and 2

Following is the titration curve of two weak monoprotic acids HA and HB (5 milli-moles each) in 50 ml aqueous solution titrated against aqueous solution of strong base NaOH (0.1 M) in two different experiments : [Given : $\log 2 = 0.3$]



1. What is pK_a value for acid HB ?
2. pH at equivalence point if HA is titrated by NaOH in the experiment given above :

Paragraph for Questions 3 and 4



3. Total number of orbitals used in bonding in anion of compounds X and Y are :
4. The pink solution and the colourless solution in the above scheme, combined to give a product in presence of ZnO catalyst. The oxidation number of central atom in the metal containing product is :

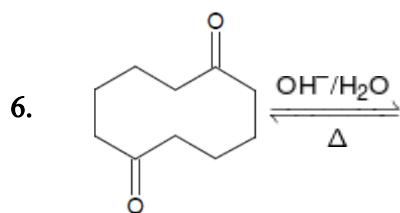
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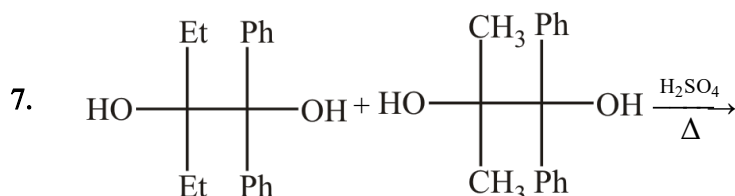
Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

5. The total number of cyclic, structural isomers for a compound with molecular formula C_5H_{10} is _____



Double bond equivalent of final product is _____



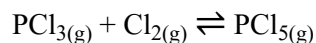
How many products are obtained in the above reaction.

8. A Hydrogen electrode filled with hydrogen gas at a pressure of 1 atm was dipped in a solution of weak monoprotic acid (HA), 20 mL of a NaOH is added and connected to a reference electrode $[Ag/AgCl/Cl^- (0.1 M) \text{ electrode}]$. E.m.f. of the cell was found to be 0.47 V. To the resulting solution, 30 mL of NaOH of the same strength was further added and new e.m.f. was found to be 0.53 V. What will be the pK_a value for weak monoprotic acid (HA) ?

$$\left[E_{Cl^-/AgCl/Ag}^0 = 0.23V, 2.303 \frac{RT}{F} = 0.06, \log 2 = 0.3 \right]$$

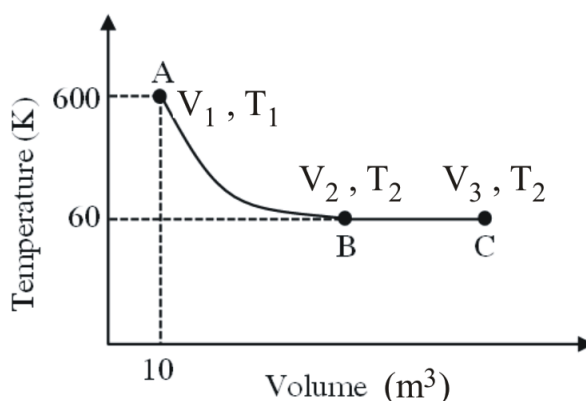
(Round off your answer to nearest integer)

9. At what temperature (in Kelvin) is the following reaction at equilibrium, when all substances are at standard pressure? Assuming that entropies and enthalpies of reaction do not vary with temperature.



	ΔH_f° (kJ/mole)	S° (J/mole-K)
$\text{PCl}_{3(g)}$	-290	310
$\text{Cl}_{2(g)}$	0	220
$\text{PCl}_{5(g)}$	-375	360

10. One mole of an ideal monoatomic gas undergoes two reversible processes ($A \rightarrow B$ and $B \rightarrow C$) as shown in the given figure :



$A \rightarrow B$ is an adiabatic process. If the total heat absorbed in the entire process ($A \rightarrow B$ and $B \rightarrow C$) is $(RT_2 \ln 10)$, then value of $(2 \log V_3)$ is _____.

[Use, molar heat capacity of the gas at constant pressure, $C_{p,m} = \frac{5}{2} R$]

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

- If x_1, x_2 and x_3 are the positive roots of the equation $x^3 - 6x^2 + 3px - 2p = 0$, $p \in \mathbb{R} - \{0\}$ then the value $\sin^{-1} \left(\frac{1}{x_1} + \frac{1}{x_2} \right) + \cos^{-1} \left(\frac{1}{x_2} + \frac{1}{x_3} \right) - \tan^{-1} \left(\frac{1}{x_3} + \frac{1}{x_1} \right)$ is equal to
 (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$
 (C) $\frac{3\pi}{4}$ (D) π
- $\sum_{r=0}^{10} 2^r {}^{20}C_r {}^{20-r}C_{10-r}$ is equal to
 (A) ${}^{20}C_{10}$ (B) ${}^{20}C_{10} \left(\frac{3}{2} \right)^{10}$
 (C) ${}^{20}C_{10} 3^{10}$ (D) ${}^{20}C_{10} 2^{10}$
- The solution of $\cos \left(\frac{x}{y} \right) (ydx - xdy) = xy^3(xdy + ydx)$, is
 (A) $\cos \left(\frac{x}{y} \right) = \frac{(xy)^2}{2} + C$ (B) $\sin \left(\frac{x}{y} \right) = \frac{(xy)^2}{2} + C$
 (C) $\cos \left(\frac{x}{y} \right) = \frac{x^2y}{2} + C$ (D) $\cos \left(\frac{x}{y} \right) = \frac{xy^2}{2} + C$
- If $\lim_{x \rightarrow 1} \sec^{-1} \left(\frac{k}{\ell \ln x} - \frac{k}{x-1} \right) = \frac{\pi}{3}$, then the value of k is equal to
 (A) 2 (B) 3 (C) 4 (D) 6

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. If the equation $\sec x + \operatorname{cosec} x = k$, ($k \in \mathbb{R}$) has exactly 3 distinct solutions between 0 and π , then k can take values
- (A) $\sqrt{2}$ (B) 2 (C) $3\sqrt{2}$ (D) 4
6. If the equation $x^4 - 4x^3 + ax^2 + bx + 1 = 0$ has four positive roots then
- (A) the eccentricity of $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $\frac{\sqrt{5}}{3}$
- (B) the eccentricity of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $\frac{\sqrt{13}}{3}$
- (C) the latus rectum of $y^2 = 4ax$ is 12
- (D) the latus rectum of $x^2 = 4by$ is 24
7. The focus of parabola is (2, 3) and it is touching the coordinate axes. Then
- (A) equation of axis of parabola is $3x - 2y = 0$
- (B) length of latus rectum of parabola is $\frac{24}{\sqrt{13}}$
- (C) equation of tangent at vertex is $3x + 2y - 6 = 0$
- (D) equation of directrix is $3x + 2y = 0$

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **TWO (02)** paragraphs.
- Based on each paragraph, There are **TWO (02)** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Consider $f(x) = x^4 + ax^3 + 3x^2 + bx + 2$ and $g(x) = x^4 + 2x^3 - bx^2 + 3x + c$, where $a, b, c \in \mathbb{R}$, and $\lim_{x \rightarrow \infty} (\sqrt{f(x)} - \sqrt{g(x)}) = 4$.

1. If a, b, c are three consecutive terms of HP. then c equals
2. If $f(x) - g(x) > 1, \forall x \in \mathbb{R}$ then number of positive integral value of c is

Paragraph for Questions 3 and 4

10 digit numbers are formed by using all the digits 0, 1, 2, ..., 9 such that they are divisible by 11111. Then, answer the following questions.

3. The digit in tens place, in smallest of such numbers is
4. The digit in units place, in greatest of such numbers is

SECTION-II (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

5. If $(1 + ax + bx^2)^4 = a_0 + a_1x + a_2x^2 + \dots + a_8x^8$; where $a, b, a_0, a_1, \dots, a_8 \in \mathbb{R}$ and $a_0 + a_1 + a_2 \neq 0$ and $\begin{vmatrix} a_0 & a_1 & a_2 \\ a_1 & a_2 & a_0 \\ a_2 & a_0 & a_1 \end{vmatrix} = 0$ then value of $\frac{5a}{b}$ is equal to
6. The line $x + 2y + c = 0$ intersects the circle $x^2 + y^2 - 4 = 0$ at two distinct points P and Q. Another line $12x - 6y = 41$ intersects the circle $x^2 + y^2 - 4x - 2y + 1 = 0$ at two distinct points R and S. Find the value of c for which the four points P, Q, R and S are concyclic.
7. The value of $\lim_{n \rightarrow \infty} \left(\sum_{r=1}^n \left(\frac{2r+5}{r^2+r} \right) \left(\frac{3}{5} \right)^{r+1} \right)$ is equal to L, then the value of 10L is
8. If the maximum value of $\left| \sqrt{x^4 - 7x^2 - 4x + 20} - \sqrt{x^4 + 9x^2 + 16} \right|$ is P, find the value of $\left\lceil \frac{P}{2} \right\rceil$ (where $\lceil . \rceil$ denotes GIF)
9. Minimum distance between the parabolas $y^2 - 4x - 8y + 40 = 0$ and $x^2 - 8x - 4y + 40 = 0$, is 'd' then $\frac{5d}{\sqrt{2}}$ is equal to
10. If $g : [1, \infty) \rightarrow [2, \infty)$ be a differentiable function and satisfying $6 \int_1^x g(t) dt = 3xg(x) - x^3 - 5 \forall x \geq 1$, then value of $\frac{1}{2}g(5)$ is equal to :

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work

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E-24/24

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